



4.3.5 Wrap-Up: Reflection

Color in the arrow up to the statement that best describes your current understanding of identifying key features of a linear function.



I'm so confident, I can explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

Unit 4, Lesson 4: Relating Key Features of Lines to Real-World Contexts



Benchmark

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.



Learning Targets

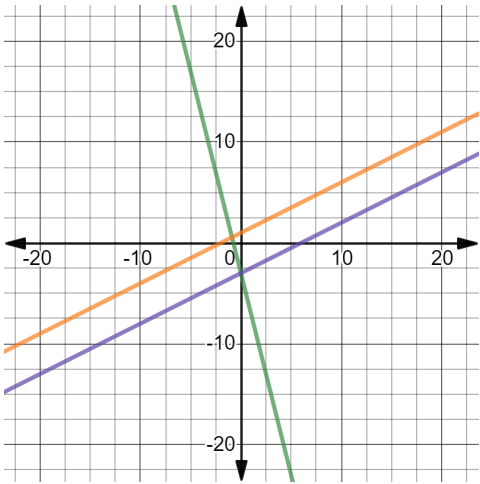
- ☐ I can locate the key features of linear functions using graphs, tables, equations, and written descriptions.
- ☐ I can determine if the domain and range are viable for the context.
- ☐ I can interpret the slope and y -intercept based on the context and determine if the interpretations are viable.



4.4.1 Warm-Up: Matching

1. Match the equations, tables, and graphs noting any key features you use to determine the matches.

	Equations
A	$4x + y = -3$
B	$y = \frac{1}{2}x - 3$
C	$y - 4 = \frac{1}{2}(x - 6)$



	Tables						
1	x	-3	-2	-1	0	1	2
	y	-4.5	-4	-3.5	-3	-2.5	-2
2	x	-3	-2	-1	0	1	2
	y	9	5	1	-3	-7	-11
3	x	-3	-2	-1	0	1	2
	y	-0.5	0	0.5	1	1.5	2

Equation	Table	Color of Graph	Key Features I Used
A			
B			
C			



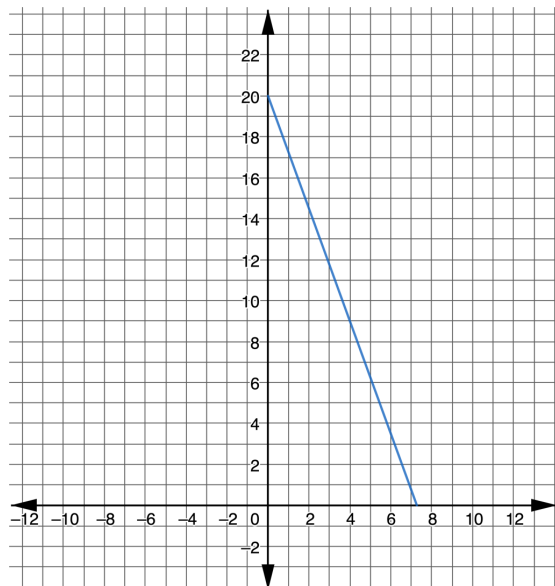
4.4.2 Collaboration: Key Features From a Graph

1. With your partner, discuss which key features they used for each graph and why they used those key features to determine the equation and table that matched the graph. Record your summary of their key features and reason. You are the only person who should write in the middle column of the table. Have your partner initial next to your summary stating that your summary was correct.

Graph	My Summary of My Partner's Key Features	Initials
Orange		
Green		
Purple		



2. Bella is using a gift card to buy coffee at a coffee shop. The initial value of the gift card is \$20. The function $A(c) = 20 - 2.75c$ represents the amount of money, A , in dollars, that is still left on the gift card after purchasing c cups of coffee.



c	A
0	20
1	17.25
2	14.5
3	11.75
4	9
5	6.25
6	3.5
7	0.75
$7.\overline{27}$	0

Work with your partner to identify the key features of the function $A(c) = 20 - 2.75c$.

Domain	Range	As x -values increase, y -values ...
Slope	x -intercept	y -intercept



The **domain** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.

The definition of the domain states, "The domain may vary depending on the context." The domain of the function $A(c) = 20 - 2.75c$ should only be whole numbers 7 or less.

3. Work with your partner to write an explanation for why the domain should only be whole numbers 7 or less. Then, if necessary, fix your domain for $A(c) = 20 - 2.75c$ in the previous problem.
4. Complete the interpretation of the slope of the function.

For every
☐ 1 cup
☐ 2.75 cups
 of coffee purchased, the

amount of money still left on the gift card

☐ decreases
☐ increases

 by

☐ \$1.00.
☐ \$2.75.

5. Work with your partner to explain what the y -intercept indicates about the gift card.



A key feature is **viable** if the value(s) are feasible for the real-world context.

6. The x -intercept for the function $A(c) = 20 - 2.75c$ is not viable for the context. Explain why the x -intercept does not make sense for the context.



4.4.3 Activity: Gallery Walk

There are four contexts shown. Use the description of the context and representations of the function to determine if each key feature on your recording sheet is viable in context.

Context	Description	Representations
A	The percentage, P , of U.S. companies who performed drug tests on employees or job applicants for x , the year since 1998, for years 1998-2003	$P(x) = -1.7x + 74.5$
B	The length of a woman's femur, f , in inches, as a function of her height, h , in inches	$f(h) = 0.432h - 10.44$

C	Sarah's parents give her a \$100 allowance at the beginning of each month. Sarah spends her allowance on comic books. The amount of allowance Sarah has remaining, y , in dollars, is a function of the number of comic books, x , that she buys for \$5 each.	$y = -5x + 100$	
		Number of Comic Books Purchased	Allowance Remaining
		2	\$90
		9	\$55
		13	\$35
		20	\$0
		24	-\$25
D	A construction company charges \$8 per hour for debris removal plus a one-time fee of \$150 for the use of a trash dumpster. The total fee, y , in dollars, is a function of the number of hours, x , the company takes to remove the debris.	$8x - y = -150$	
		Number of Hours	Fee (\$)
		-10	\$70
		-4	\$118
		0	\$150
		5	\$190
		24	\$342

Gallery Walk Recording Sheet

Use the recording sheet below to record whether each key feature is viable for the context. Include any notes that will help you remember why.

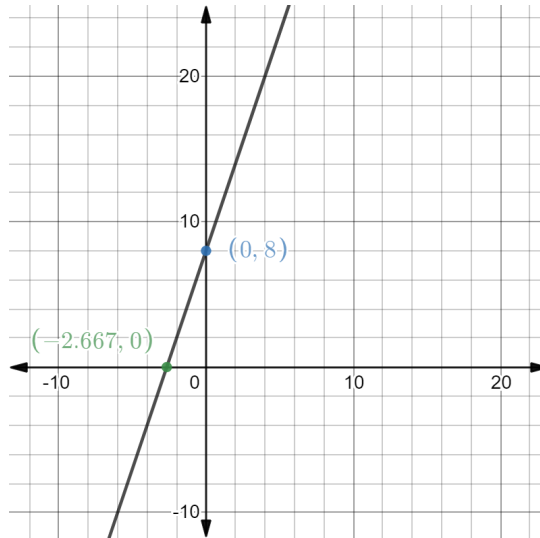
Context	Domain	Range	x -intercept	y -intercept
A				
B				
C				
D				



4.4.4 Wrap-Up: Ticket Out the Door

Antonella places x marbles into an empty container. Each marble has the same weight.

The weight, W , in ounces, of the container and the marbles can be found using the equation $W(x) = 3x + 8$.



- Li Qiang completed the table of four key features. Determine if the key features given by Li Qiang are viable. If the given key feature is not viable for the context, explain why it is NOT viable.

	Domain	Range	x -intercept	y -intercept
Key Feature	$x \geq 0$	$y \geq 0$	$(-\frac{8}{3}, 0)$	$(0, 8)$
Viable? Yes/No				
Explanation				



Unit 4, Lesson 5: Graphing and Interpreting Key Features of Lines



Benchmark

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.



Learning Target

- ☐ I can graph a line and interpret its key features.